The Integrity Protocol

Recursive Neural Trust & Entropy-Aware Reputation for the Agentic Web

1. Abstract: The Thermodynamics of Truth

The Integrity Protocol introduces a model of **Biological Trust**, grounded in the physical reality of computational work and technical order. By anchoring reputation in energy expenditure and behavioral stability, we create a trust layer that is physically and mathematically expensive to fake.

2. The Tri-Metric Derivations

2.1 Entropy Score (S_e)

Technical stability is the foundation of trust. We analyze the variance (σ^2) across an agent's latency and accuracy streams.

$$S_e = 1000 \cdot e^{-1.5 \cdot \mathrm{CV}}$$

2.2 Grounding Score (S_g)

The Grounding Score measures human accountability via the **Grounding Delta**—the temporal and semantic signature of human-in-the-loop oversight.

$$S_g = (Ratio_{intervention} \cdot 0.4 + Depth_{edit} \cdot 0.6) \cdot 1000$$

2.3 Integrity Score (S_i)

The sovereign reputation backed by **Computational Sacrifice** (C_i).

$$S_i = [(T_i \cdot 0.25) + (X_i \cdot 0.25) + (C_i \cdot 0.50)] \cdot \text{Stability_Drag}$$

3. Tokenomics: INTG

INTG is the ERC-20 utility token powering the ecosystem. It is used for staking, slashing, and settlement. 50% of every 0.5% verification fee is permanently burned, creating a deflationary asset class tied to the growth of AI trust.

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